Development summary: stick-knot counting

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Manuscript authors: Alexander Kolpakov and Igor Rivin. This retrospective account summarizes released mathematical sources and the corpus-preparation audit. It is not a transcript or a reconstruction of model turns. No original conversation export was available in the source tree.

Problem and result

The mathematical problem is to count ordinary knot types that can be built with at most N straight sticks. For their set $\mathcal{K}_N$, the manuscript proves

$$N^{(2/3+o(1))N} \leq |\mathcal{K}_N| \leq N^{3N+o(N)}.$$

Thus $\log |\mathcal{K}_N| = \Theta(N \log N)$. This is a sharp logarithmic growth order, not an asymptotic equivalent or a determined leading constant.

Mathematical route

1. Represent an N-vertex polygon by $3N$ real coordinates. A nonincident-edge intersection forces a cubic determinant to vanish; quartic squared cross products exclude collapsed or overlapping adjacent edges. There are only quadratically many walls.

2. Delete all these walls, including benign degeneracies. Every bounded-stick knot has a representative in the resulting open locus by subdivision and generic perturbation. Knot type is constant along paths, so a surjection from components bounds the knot population.

3. Apply the refined sign-condition component estimate with its partial binomial sum retained. Dividing by the dimension factorial changes the relevant base from quadratic to linear in N, giving the upper exponent 3.

4. Construct doubly down–up permutations on $4k$ letters from four independent permutations on k letters. The family has at least $(k!)^4$ elements and logarithmic deficit only $O(k)$ relative to $(4k)!$.

5. Equal starting and finishing sets make arbitrary words in these simple braids distinct normal forms. Pigeonhole by endpoint permutation and append one common inverse-endpoint factor. The pure braids remain distinct and use at most one additional simple factor.

6. Transfer through the Malyutin–Stupakov injection and Huh–Oh arc-to-stick inequality, losing at most a factor two on forgetting orientation. Fixed word lengths give every coefficient below $2/3$; word length $\lfloor \log N \rfloor$ yields the more precise lower deficit $-(2/3)N \log\log N - O(N)$.

Priority and audit decisions

To the authors' knowledge, the manuscript gives the first explicit determination of the logarithmic growth order in the stick budget. Older arc-index results supply the lower-bound inputs. The 2025 strong-geometry reconstruction theorem is related work that could, in principle, supply another counting route after additional deductions. The present proof uses direct component counting independently of that reconstruction theorem. The novelty audit retains a separate retrospective comparison, clearly identified as a deduction made in this project.

The corpus audit checked the external statements used at the main interfaces and the manuscript's internal counting and budget calculations. It made the eventual positivity of the fixed-word strand count explicit. It also supplied a concrete circle-and-return-rail routing model to justify the uniform stick cost in the auxiliary addressed-link construction.

Formal verification

The standalone Lean project uses actual polygon coordinates, multivariate wall polynomials, paths and component quotients, permutation words, endpoint fibers and a bounded-stick subtype of a supplied knot interface. It checks their internal finite and asymptotic deductions. The corpus preparation adds a proof of the previously separate optimized floor/log arithmetic and an eventual upper coefficient theorem for every coefficient above 3.

Generic support, ambient-isotopy invariance, Barone–Basu, the structural Garside facts, the pure-braid-to-knot injection, the arc-to-stick inequality and orientation reversal remain explicit external theorem inputs. The classical ambient-isotopy quotient and the auxiliary link construction are not formalized. A successful axiom audit verifies these conditional proofs; it does not prove their theorem parameters.

Source coverage

Sources are the manuscript, novelty audit, formal-verification documentation, the starting publication history and the new mathematical audit. They support this mathematical account, not an authenticated chronological conversation. The source acknowledgments report GPT-5.6-sol assistance; the underlying model run, original prompt and compute expenditure are not authenticated here. Public authorship is retained. Private conversation records, credentials, account/session fields and local machine paths are not included.